

TASK MANAGEMENT SYSTEM

TIME COMPLEXITY:

Operation	Time Complexity	Explanation
Add Task	$O(n)$	Traverse to the last node and insert the new task.
Search Task	$O(n)$	May need to check every node until the task is found.
Traverse Tasks	$O(n)$	Visits each node exactly once.
Delete Task	$O(n)$	First search for the node, then remove it.

ADVANTAGES OF LINKED LISTS OVER ARRAYS FOR DYNAMIC DATA:

- Dynamic Size
- Efficient Insertion and Deletion
- No Continuous Memory Requirement
- Better Memory Utilization
- Easy Implementation of Dynamic Data Structures